

llm_task2_2_Llama32_1B_buyer_agentic_workflo

February 5, 2026

0.1 Buyer agentic workflow pipeline

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[12]: # 1. Imports and LLM configuration
from typing import Dict, Any # <- needed for type hints

import os
import pandas as pd
import numpy as np
import joblib
import requests
import textwrap

LM_STUDIO_URL = "http://localhost:1234/v1/chat/completions"
LM_MODEL_NAME = "llama-3.2-1b-instruct" # matches the API identifier in LM_
↳Studio

# 2. Load the enhanced dataset (with sentiment scores)
current_dir = os.path.dirname(os.path.abspath("__file__")) if "__file__" in_
↳globals() else os.getcwd()
output_dir = os.path.join(current_dir, "output")

data_file = os.path.join(output_dir, "df_with_extracted_scores.csv")
df = pd.read_csv(data_file)
print("Loaded data:", data_file)
print("Columns:", df.columns.tolist())

# 3. LLM helper for buyer assistant (reuses the same API as seller notebook)
def call_llm(prompt: str) -> str:
    """
    Call LLaMA-3.2-1B-Instruct via LM Studio's local OpenAI-compatible API.
    Make sure LM Studio server is running (Status: Running).
    """
    payload = {
        "model": LM_MODEL_NAME,
        "messages": [
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        {"role": "system", "content": "You are a helpful real-estate_
↪assistant."},
        {"role": "user", "content": prompt},
    ],
    "temperature": 0.3,
    "max_tokens": 512,
}

try:
    resp = requests.post(LM_STUDIO_URL, json=payload, timeout=120)
    resp.raise_for_status()
    data = resp.json()
    return data["choices"][0]["message"]["content"]
except Exception as e:
    print("Error calling LLM:", e)
    return "Error: LLM call failed."

# 4. Buyer preference structure (input config)
def default_buyer_preferences() -> Dict[str, Any]:
    """
    Example buyer preferences for testing.
    """
    return {
        # Budget wide enough to include ~£25M luxury properties
        "budget_min": 20_000_000,
        "budget_max": 30_000_000,

        # Match large luxury family homes
        "min_bedrooms": 6,
        "min_bathrooms": 6,
        "min_size_sqft": 10_000,

        "preferred_property_types": ["House"],

        # Emphasise luxury and good transport/schools
        "weight_luxury": 0.5,
        "weight_transport": 0.25,
        "weight_school": 0.25,
        "weight_budget_closeness": 0.2,

        # Show a few top candidates around that segment
        "top_k": 5,
    }

# 5. Filter + ranking for buyer preferences (data-side logic)
def filter_properties_for_buyer(df: pd.DataFrame,

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                                prefs: Dict[str, Any]) -> pd.DataFrame:
    """
    Filter the full listings DataFrame according to buyer preferences.
    Assumes df has columns: price, bedrooms, bathrooms, sizeSqFeetMax,
    propertyType, luxury_score, transport_score, school_score, renovation_score.
    """
    filtered = df.copy()

    # Numeric filters
    if "budget_min" in prefs:
        filtered = filtered[filtered["price"] >= prefs["budget_min"]]
    if "budget_max" in prefs:
        filtered = filtered[filtered["price"] <= prefs["budget_max"]]

    if "min_bedrooms" in prefs:
        filtered = filtered[filtered["bedrooms"] >= prefs["min_bedrooms"]]
    if "min_bathrooms" in prefs:
        filtered = filtered[filtered["bathrooms"] >= prefs["min_bathrooms"]]
    if "min_size_sqft" in prefs:
        filtered = filtered[filtered["sizeSqFeetMax"] >= prefs["min_size_sqft"]]

    # Categorical filter: propertyType
    if prefs.get("preferred_property_types"):
        filtered = filtered[filtered["propertyType"].
↪isin(prefs["preferred_property_types"])]

    return filtered

def add_matching_score(filtered: pd.DataFrame,
                        prefs: Dict[str, Any]) -> pd.DataFrame:
    """
    Add a 'matching_score' column representing how well each property
    matches the buyer's preferences (simple heuristic scoring).
    """
    df_scored = filtered.copy()

    # Normalised sentiment contributions (0-1)
    df_scored["luxury_match"] = df_scored["luxury_score"] / 2.0 # 0-2 -> 0-1
    df_scored["transport_match"] = df_scored["transport_score"] # 0 or 1
    df_scored["school_match"] = df_scored["school_score"] # 0 or 1

    w_lux = prefs.get("weight_luxury", 0.4)
    w_tr = prefs.get("weight_transport", 0.3)
    w_sc = prefs.get("weight_school", 0.3)

    sentiment_score = (

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        w_lux * df_scored["luxury_match"]
        + w_tr * df_scored["transport_match"]
        + w_sc * df_scored["school_match"]
    )

    # Budget closeness: 1.0 if near the middle of the budget range
    mid_budget = 0.5 * (prefs["budget_min"] + prefs["budget_max"])
    max_dev = 0.5 * (prefs["budget_max"] - prefs["budget_min"]) + 1e-9
    budget_dev = (df_scored["price"] - mid_budget).abs() / max_dev
    budget_closeness = 1.0 - budget_dev.clip(0, 1)

    w_budget = prefs.get("weight_budget_closeness", 0.2)

    df_scored["matching_score"] = sentiment_score + w_budget * budget_closeness

    return df_scored

# 6. Build LLM prompt for buyer recommendations
def build_buyer_prompt(prefs: Dict[str, Any],
                      candidates: pd.DataFrame) -> str:
    """
    Build a compact text prompt describing buyer preferences and
    top candidate properties for LLaMA-3.2-1B-Instruct.
    """
    # Limit number of candidates to keep prompt short
    top_k = prefs.get("top_k", 5)
    subset = candidates.head(top_k).copy()

    rows_text = []
    for idx, row in subset.reset_index(drop=True).iterrows():
        pid = f"P{idx+1}" # simple property ID
        rows_text.append(
            f"{pid}: price=${row['price']:, .0f}, "
            f"type={row['propertyType']}, "
            f"size={row['sizeSqFeetMax']} sqft, "
            f"beds={row['bedrooms']}, baths={row['bathrooms']}, "
            f"luxury_score={row['luxury_score']}, "
            f"transport_score={row['transport_score']}, "
            f"school_score={row['school_score']}, "
            f"renovation_score={row['renovation_score']}"
        )

    properties_block = "\n".join(rows_text) if rows_text else "No properties_
    ↪found."

    prefs_text = f"""

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Budget range: £{prefs['budget_min']:, .0f} - £{prefs['budget_max']:, .0f}
Minimum bedrooms: {prefs['min_bedrooms']}
Minimum bathrooms: {prefs['min_bathrooms']}
Minimum size: {prefs['min_size_sqft']} sqft
Preferred property types: {"", ".join(prefs['preferred_property_types'])}
"""

    instructions = f"""
You are an assistant helping a home buyer choose suitable properties.

Buyer preferences:
{prefs_text.strip()}

Candidate properties (each with an ID):
{properties_block}

Task:
- You MUST ONLY refer to properties by the IDs listed above (P1, P2, P3, etc.).
  Never invent new IDs or mention properties like P5, P6, etc. if they do not
  appear in the list.
- If there are no suitable properties, explain this politely and suggest how
  the buyer could adjust their criteria.
- Otherwise:
  1) Identify the single best matching property ID from the list.
  2) Suggest 1-2 additional alternatives from the SAME list.
  3) Explain in a short paragraph why these properties fit the buyer's needs,
     focusing on budget, size, bedrooms, and the sentiment scores
     (luxury_score, transport_score, school_score, renovation_score).
- Keep the style clear and conversational.
"""

    return instructions.strip()

def recommend_properties_with_llm(prefs: Dict[str, Any],
                                  candidates: pd.DataFrame) -> str:
    """
    Wrapper that builds the buyer prompt and sends it to the LLM.
    """
    prompt = build_buyer_prompt(prefs, candidates)
    return call_llm(prompt)

# 7. End-to-end buyer agent function (agentic workflow)
def buyer_assistant(
    buyer_prefs: Dict[str, Any]
) -> Dict[str, Any]:

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"""
High-level agentic workflow for a buyer.

Steps:
1) Filter the full dataset according to buyer preferences.
2) Compute a matching_score for each remaining property.
3) Sort by matching_score and select top-k candidates.
4) Ask the LLM to recommend and explain best matches.

Returns:
- filtered_ranked: DataFrame of candidate properties with matching_score.
- llm_buyer_advice: string with the LLM's recommendations/explanations.
"""

# 1) Filter
filtered = filter_properties_for_buyer(df, buyer_prefs)

if filtered.empty:
    # Still ask LLM to explain "no matches", for consistency
    empty_df = filtered.copy()
    llm_output = recommend_properties_with_llm(buyer_prefs, empty_df)
    return {
        "filtered_ranked": empty_df,
        "llm_buyer_advice": llm_output,
    }

# 2) Add matching score
scored = add_matching_score(filtered, buyer_prefs)

# 3) Sort by matching_score (descending)
scored_sorted = scored.sort_values(by="matching_score", ascending=False)

# 4) Ask LLM
llm_output = recommend_properties_with_llm(buyer_prefs, scored_sorted)

return {
    "filtered_ranked": scored_sorted,
    "llm_buyer_advice": llm_output,
}

```

Loaded data:

```

c:\Users\Admin\Python\S8_Thesis\llm\output\df_with_extracted_scores.csv
Columns: ['title', 'propertyType', 'sizeSqFeetMax', 'bedrooms', 'bathrooms',
'listingUpdateReason', 'price', 'Date', 'listingDescription', 'luxury_score',
'transport_score', 'school_score', 'renovation_score']

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[13]: # 8. Example usage (UI-style display for buyer, for thesis screenshots)
def display_buyer_session(buyer_prefs: Dict[str, Any], result: Dict[str, Any]):

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print("=" * 80)
print("BUYER ASSISTANT - RECOMMENDATIONS".center(80))
print("=" * 80)

print("\n[Buyer preferences]")
print(f" Budget range           : &{buyer_prefs['budget_min']:, .0f} - "
      f"&{buyer_prefs['budget_max']:, .0f}")
print(f" Min bedrooms           : {buyer_prefs['min_bedrooms']}")
print(f" Min bathrooms          : {buyer_prefs['min_bathrooms']}")
print(f" Min size (sq ft)       : {buyer_prefs['min_size_sqft']}")
print(
    " Preferred property types: "
    f"{', '.join(buyer_prefs['preferred_property_types'])}"
)

print("\n" + "-" * 80)
print("TOP MATCHING PROPERTIES (internal view)".center(80))
print("-" * 80)

candidates = result["filtered_ranked"].head(buyer_prefs.get("top_k", 5))
if candidates.empty:
    print("No properties found that satisfy these filters.")
else:
    for i, (_, row) in enumerate(candidates.iterrows(), start=1):
        print(f"\n[Property P{i}]")
        print(f" Price           : &{row['price']:, .0f}")
        print(f" Type           : {row['propertyType']}")
        print(f" Size (sq ft)    : {row['sizeSqFeetMax']}")
        print(f" Bedrooms       : {row['bedrooms']}")
        print(f" Bathrooms      : {row['bathrooms']}")
        print(f" luxury_score    : {row['luxury_score']}")
        print(f" transport_score : {row['transport_score']}")
        print(f" school_score    : {row['school_score']}")
        print(f" renovation_score: {row['renovation_score']}")
        print(f" matching_score  : {row['matching_score']:.3f}")

print("\n" + "-" * 80)
print("LLM RECOMMENDATION SUMMARY".center(80))
print("-" * 80)

advice = result["llm_buyer_advice"]
print(textwrap.fill(advice, width=80))

print("\n" + "=" * 80)

```

9. Run one example buyer session

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example_buyer_prefs = default_buyer_preferences()
buyer_result = buyer_assistant(example_buyer_prefs)
display_buyer_session(example_buyer_prefs, buyer_result)
```

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BUYER ASSISTANT - RECOMMENDATIONS

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[Buyer preferences]

Budget range : £20,000,000 - £30,000,000
Min bedrooms : 6
Min bathrooms : 6
Min size (sq ft) : 10000
Preferred property types: House

TOP MATCHING PROPERTIES (internal view)

[Property P1]

Price : £24,950,000
Type : House
Size (sq ft) : 16749
Bedrooms : 8.0
Bathrooms : 8.0
luxury_score : 2
transport_score : 1
school_score : 1
renovation_score: 0
matching_score : 1.198

[Property P2]

Price : £29,500,000
Type : House
Size (sq ft) : 12960
Bedrooms : 7.0
Bathrooms : 7.0
luxury_score : 1
transport_score : 1
school_score : 0
renovation_score: 0
matching_score : 0.520

[Property P3]

Price : £29,500,000
Type : House
Size (sq ft) : 12960
Bedrooms : 7.0

Bathrooms : 7.0
luxury_score : 1
transport_score : 0
school_score : 0
renovation_score: 0
matching_score : 0.270

[Property P4]

Price : £29,950,000
Type : House
Size (sq ft) : 15608
Bedrooms : 6.0
Bathrooms : 6.0
luxury_score : 0
transport_score : 1
school_score : 0
renovation_score: 0
matching_score : 0.252

LLM RECOMMENDATION SUMMARY

Hi there! I'd be happy to help you find a suitable property. First, let's review your criteria: £20,000,000 - £30,000,000 budget, minimum of 6 bedrooms, 6 bathrooms, and a minimum size of 10,000 sqft. You also prefer houses over other types of properties. Now, let's take a look at the properties in our list: P1 (£24,950,000) - House, 16749 sqft, 8 beds, 8 baths P2 (£29,500,000) - House, 12960 sqft, 7 beds, 7 baths P3 (£29,500,000) - House, 12960 sqft, 7 beds, 7 baths P4 (£29,950,000) - House, 15608 sqft, 6 beds, 6 baths After reviewing these properties, I think the single best matching property ID is P1. Here's why: * The price of £24,950,000 falls within your budget range. * The house has a good balance of size (10,500 sqft) and bedrooms (8). * The number of bathrooms remains at 8, which meets your requirement. As for additional alternatives, I'd suggest considering properties with slightly higher prices that still meet the minimum requirements. P2 (£29,500,000) is another option to consider, as it offers a similar size and bedroom count but has a slightly lower price point than P1. In terms of sentiment scores, all four properties have a score of 0 for luxury, transport, school, and renovation. This suggests that none of these properties are particularly luxurious, convenient, or well-regarded in the schools they attend. Overall, I think P1 (£24,950,000) is an excellent choice given your criteria.